Reciprocal relationships, reverse causality, and temporal ordering: Testing theories with cross-lagged panel models

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Reciprocal causal relationships are a common feature of criminological theories. For example, police forces tend to use force more often in areas where crime concentrates, while at the same time legal cynicism theory suggests that cumulative exposures to police use-of-force can foster criminal activity. When multiple observations over time are available, cross-lagged panel models are commonly used to estimate these reciprocal effects. This is often done without careful attention to the assumptions that must be satisfied to produce valid estimates, such as correctly specified temporal lags, sufficient inter-temporal variation, and proper accounting for unobserved heterogeneity. Failure to satisfy these assumptions can produce severe issues including spurious associations and parameter estimates that are biased or even reversed in direction. In addition, reciprocal relationships violate causal assumptions based on graphical tools; criminological theories that suggest reciprocal causal relationships usually have an underlying macro-micro mechanism often not accounted for in empirical models. We provide guidance on how to align theory, model specification, and choice of estimator and illustrate this using familiar topics in life course research. We conclude by highlighting the importance of criminological theory and careful attention to empirical implications of theoretical premises when investigating reciprocal relationships

Introduction

Reciprocal relationships and reverse causality are a common feature of criminological theories. In some cases, theories suggest that two variables are reciprocally related, reproducing each other over time in a vicious cycle. For example, labeling theory suggests individuals labelled as criminals are more likely to engage in criminal activity, which then contributes to reinforcing their labels as criminals, forming a cyclic pattern of criminal reproduction (Becker 1963). In other cases, competing theories suggest distinct explanations regarding the causal direction of an established association. For example, social disorganization theories suggest social and physical disorder in the neighborhood increases people's fear of crime, whereas socio-psychological theories suggest that fear of crime contributes to heightened perceptions of neighborhood disorder (Sampson and Raudenbush 2004; Jackson 2004). In still other cases, theories suggest a causal effect of a treatment on an outcome, while also cautioning that the outcome can affect the treatment over time. For example, developmental and life-course criminology suggests that stable employment reduces criminal behavior, whereas criminal behavior can sometimes also lead to job loss (Ramakers et al. 2017; **?**). Across all these examples, reciprocality—the possibility that two variables influence each other over time—is at the heart of the theoretical process.

When repeated observations over time are available, the cross-lagged panel model (CLPM) is one of the most commonly used tools for studying these reciprocal processes. In its simplest form, a CLPM regresses each variable on its own lagged value and on the lagged value of the other variable, producing autoregressive and cross-lagged parameter estimates. A credible non-zero cross-lagged coefficient is typically interpreted as evidence that one variable influences the other over time, net of their respective stabilities. Because both cross-lagged paths can be estimated simultaneously, the CLPM appears to offer a convenient way to assess reciprocality or to adjudicate between competing causal directions—and it has been applied to a wide range of topics in criminology and beyond (Brunton-Smith 2011; Lanfear and Kirk 2024; Pina-Sánchez and Brunton-Smith 2020).

Yet CLPMs are frequently used without careful attention to the theoretical process they are meant to capture. In particular, studies sometimes fail to consider the role of time when specifying temporal processes. Causal processes unfold at particular speeds—for example, while employment may constrain offending on a daily basis through routine activities, the reputational consequences of a criminal record may take months to translate into job loss [REFS]. These temporal dynamics should inform not only modelling strategies but, more importantly, the definition of the causal quantity of interest—the *estimand*, which is often left implicit, determined not by theory but by the structure of the available data. When the starting point is not a clearly specified causal process, one risks arriving at an estimator that produces a correct answer to the wrong question. In addition, the apparent simplicity of the CLPM belies a number of assumptions that must be satisfied for its estimates to be valid. These include correctly specified temporal lags between exposure and outcome, sufficient within-unit variation over time to detect cross-lagged effects, and proper accounting for unobserved heterogeneity that might confound the estimated relationships. Failure to satisfy these assumptions can produce severe problems, including spurious associations, imprecise estimates, and parameter estimates that are biased or even reversed in direction.

Our goal in this paper is to provide practical guidance for researchers studying reciprocal relationships or dealing with reverse causality using panel data. We do not develop new estimators or derive new formal results, but we synthesise several recent methodological contributions that have been developed by different disciplinary traditions (e.g., Allison, Williams, and Moral-Benito 2017; Hamaker, Kuiper, and Grasman 2015; Zyphur et al. 2020) in a way that is directly accessible to applied researchers in criminology. First, we highlight the nuanced but important distinction between *theoretical reciprocality*, when feedback processes are the core of the theories being tested, and *reverse causality*, when reciprocal effects are a nuisance of the theory being tested. We focus on three key problems that we believe are the most consequential for applied research in developmental and life-course criminology: (a) improperly specified temporal lags, (b) unobserved heterogeneity, and (c) insufficient inter-temporal variation. These are not the only issues that can arise with CLPMs, but they are the ones that researchers will most frequently encounter and that can have the most severe consequences for inference. For each, we explain why it matters, how it arises in practice, and what solutions are available.

We ground our discussion in two motivating examples that represent the two broad reasons one might turn to a CLPM. The first is the relationship between employment and criminal offending—a staple of life-course criminology (Sampson and Laub 1993; Crutchfield 2014). Here, the primary research question concerns the causal effect of employment on crime, but reverse causality (crime involvement leading to job loss) threatens to bias this any attempts to estimate this effect. In this case, reciprocality is a nuisance to be controlled for. The second is the relationship between perceived neighborhood disorder and fear of crime—a central topic for neighborhood researchers (Sampson and Raudenbush 2004; O'Brien, Farrell, and Welsh 2019). Here, competing theories propose opposite causal directions, and the reciprocal relationship itself is the substantive question. We use these examples to illustrate how the same general class of models can serve different research purposes, and how the distinction between reverse causality as a nuisance and theoretical reciprocality as a substantive questions should also shape model specification.

The paper proceeds as follows. We first discuss estimands and the role of time, highlighting what we mean by reciprocal effects in the context of theory testing and the role of unexamined mechanisms operating at a lower, more temporally disaggregated level than observed measures. We then present our two motivating examples in detail, with focus on the distinction between reverse causality and theoretical reciprocality. We then develop the statistical framework and discuss why we recommend using directed acyclic graphs (DAGs) to represent the theoretical model and structural equation models (SEMs) for estimation, while being attentive to where these two representations necessarily diverge. The main body of the paper addresses the three main issues—temporal order, unobserved heterogeneity, and insufficient inter-temporal variation—as well as the available solutions for each of them. We conclude with general recommendations for researchers studying reciprocal relationships with panel data.

Estimands and the role of time

An estimand is the quantitative definition of the estimate of interest such as a causal effect (see Lundberg et al. 2021). In a counterfactual framework, an estimand may be defined as a difference in potential outcomes, e.g., "how much higher is the probability that an individual will offend in a given month if they became unemployed in the prior month?" Estimands do not exist out in the world; they are chosen by researchers. We defer to other sources for a full explanation of choosing and using estimands (e.g., Lundberg et al. 2021). For the present work, what is important is recognizing that there is no "true causal timing" waiting to be discovered because causal timing is a choice a researcher makes when defining their estimand of interest.

To illustrate, consider this: what is the true causal timing of drinking an additional pint of beer on perceived wellbeing? What is this effect at 20 seconds, 20 minutes, 20 hours, and 20 years? What about the effect of a pint per day after 10 years? In this context, the reader may know intimately which exposure and which lag(s) between measurement of exposure and outcome correspond to detectable and substantively interesting causal effects. Strong theories should be specific about the role of time and thus able to inform what sort of causal timings—whether delay between exposure and outcome or aggregations of exposures—are of substantive interest. Causal processes are often (but not always) continuous but researchers discretize time for practical utility, e.g., we know drinking a pint of beer causes your blood alcohol level to ramp up then back down but may be interested in knowing how they will feel 30 minutes from now or when they wake up in 12 hours–or how it will impact their health in ten years if they make it regular habit.

Often the estimates produced in longitudinal research do not correspond to any theoretically-motivated or substantively useful estimand. While a relatively common specification in longitudinal research, consider when we would be interested in the (lagged) effect of percent of time employed last year on crimes committed this year. Does this estimand make sense given how we think employment is related to offending? Implicit estimands such as these are often chosen by starting with a methods concern (e.g., reverse causality) and working backwards, so one arrives at a model that gives a correct answer to the wrong question.

Relatedly, researchers should recognize that because causal effects occur over time, there are no simultaneous non-recursive causal relationships (Strotz & Wold 1960). When researchers posit non-recursive relationships, they are nearly always conducting analysis at a higher-level (i.e., macro) than where the process of interest is taking place (i.e., micro). For example, while economics depicts supply and demand as simultaneously determined, this relationship becomes recursive if we make explicit the underlying micro-macro model in which individual actors on both sides adjust to each others' behavior over time to produce the equilibrium price. Abstraction to higher levels of analysis is both useful and ubiquitous throughout the sciences and formal guidelines exist for when macro-level causal models will accurately represent (or translate) the outcomes of micro-level causal models (e.g., Rubenstein et al. 2017). These abstractions are often forced on researchers by limitations in measurement. Long time lags in longitudinal research are a special case of this micro-macro issue in which faster-operating micro processes (e.g., daily processes) are a modeled at a longer macro time scale (e.g., using waves of an annual survey).

Consequently, researchers should make an argument that the timing of the effect in the estimand they are studying is useful and can be studied with the data they have. While we focus here on longitudinal research, this applies just as much to cross-sectional research, except there researchers have even less control over what timings they may examine. Even provided otherwise identical estimands, discrepant results between studies may be due to different choices of timing (Gollob & Reichardt 1987; see also Vaisey & Miles 2017); the effect of an additional pint on perceived wellbeing will be quite different measured in one hour versus tomorrow morning.

Two motivating examples

To illustrate our points, we consider two canonical longitudinal relationships in criminology. The first is the link between employment and offending—a focus of classic life-course research (e.g., Sampson and Laub 1992; Crutchfield 2014). The second is the link between residents' perceptions of neighborhood disorder and their fear of crime—a central topic for neighborhood scholars (Sampson and Raudenbush 2004; O'Brien, Farrell, and Welsh 2019). In the employment–crime example, theory and policy emphasize the effect of employment on future offending, whereas in the fear–disorder example the direction of causation is itself debated.

Employment and crime

Life-course studies routinely find that obtaining stable employment reduces criminal behavior (Ramakers et al. 2017; Lageson and Uggen 2012). The logic is straightforward. In the short run, being employed fills one's time with routine activities and conventional social bonds, which constrains opportunities and incentives for offending (Cohen and Felson 2010; Hirschi 2017). In the long run, it is not employment per se but stable, committed work that enhances social control and thereby reduces crime (Sampson and Laub 1993). In other words, the accumulation of conventional ties and responsibilities that are often associated with stable employment provides identity and that deters future offending (Ramakers et al. 2017). For example, longitudinal reviews show that employed individuals have significantly lower recidivism rates than the unemployed (Uggen and Wakefield 2008; although see Crutchfield 2014). Indeed, some experimental job-program evaluations have found that random assignment to job training can yield lower subsequent crime (Redcross et al. 2011).

Crucially, the reverse causal path is also possible. After all, criminal arrests or behavior can lead to job loss (Crutchfield 2014). Therefore, when estimating the effects on employment on crime, one needs to account for reverse causality. The crime → employment relationship is not of immediate substantive interest to most criminologists, so in this example we regard that as an endogeneity problem. Our substantive interest is in the effect of employment on later crime, treating any effects of crime involvement on employment as a nuisance and a source of bias which needs to be controlled for.

Fear of crime and perceived disorder

By contrast, the relationship between perceived disorder and fear of crime is typically viewed as reciprocal. Some posit that visible disorder in the environment raises fear of crime—disorderly cues signal that an area is in decline and unable to control deviance (Kelling, Wilson, et al. 1982). Others emphasize the opposite interpretation, that individuals with high levels of fear of crime are sensitized to local disorder, causing them to perceive and interpret interpret ambiguous cues as physical and social disorder (Sampson and Raudenbush 2004; Jackson 2004). In practice, both processes may occur (Brunton-Smith 2011). Fear and disorder perceptions may result from rapid updating processes. Given the theoretical ambiguity and the possibility for reciprocal relationships over time, this example does not have clearly defined dependent and independent variables. Both fear of crime and perceived disorder are outcomes as well as predictors of interest. Our substantive interest is therefore the reciprocality itself. We can use panel models either to adjudicate between competing theories or to discover actual reciprocal effects over time.

Theoretical and statistical models

Both examples are contrasted by how they are framed theoretically. In the employment–crime case, the cross-lagged panel model is used to account for reverse causality (crime $\rightarrow$ job) and isolate the employment $\rightarrow$ crime effect of interest. In the fear–disorder case, by contrast, both lagged paths are of substantive interest, reflecting genuine theoretical ambiguity. Which direction we emphasize is dictated by the research question. In the first example we treat the reverse path as a nuisance (i.e., reverse causality), whereas in the second we treat it as part of a mutually reciprocal process to be tested (i.e., theoretical reciprocality).

Good empirical research starts with a strong theoretical foundation. It is from a clearly defined theoretical framework that observable implications can be deduced, and it is those observable implications which can then be contrasted against empirical data (King, Keohane, and Verba 1994; Mayo 2018). Theory-testing empirical research is therefore conditional on a thorough and transparent theoretical model (Nagin and Sampson 2019). Given that most criminological theories presume causal processes, the first step should normally involve deriving a non-recursive model that accurately represents the theorized relationships—e.g., the theoretical model could be represented by a directed acyclic graph (DAG). DAGs can formally represent a nonparametric structural equation model, avoiding functional-form and distributional assumptions (Pearl 2009), and are increasingly used to portray premised causal relationships in criminological theory (Nivette and Vegt 2025; Oliveira 2024). The DAG should be all-encompassing and include all relevant theorized pathways, including unobserved mechanisms.

Once the theoretical model is specified, it should guide the selection of an appropriate estimator. It is important to distinguish the theoretical model from the statistical model: while the latter should approximate the former as much as possible, the statistical model will unavoidably have to handle imprecision and uncertainty that are bound to exist when dealing with empirical data, such as distributional and parametric assumptions and unanalysed covariances due to theoretical ambiguities. When dealing with situations involving potential reciprocal relationships and/or reverse causality, we propose leveraging DAGs for theoretical models and drawing on the structural equation modeling (SEM) framework for estimation (Bollen and Pearl 2013). A clear theoretical model is a necessary prerequisite for specifying an estimator. While both diagrams will not be exactly the same, we suggest that the premised DAG inform some elements of the SEM. Specifically, uni-directional arrows in the structural model should be non-recursive, as per the theoretical model.[1] For example, as depicted in Figure 1, a key consideration in this process is the timing of effects: theorized effects can be contemporaneous (i.e., occur quickly) or lagged (i.e., unfold more gradually over time).

(a) Lagged effects only (b) Contemporaneous effects

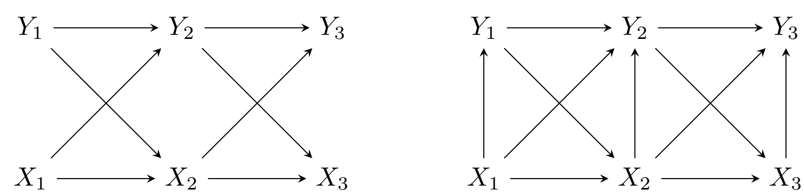

Figure 1: Directed acyclic graph representation of cross-lagged panel models with (a) lagged effects only or (b) contemporaneous effects as well

Yet, while the theoretical model displays a non-parametric representation of theorized causal relationships, the SEM can include bi-directional arrows representing empirically observed covariances between variables not explained by the premised DAG. Panel (a) in Figure 2 depicts a statistical model corresponding to the theoretical model in panel (a) of Figure 1, where the reciprocal effects of interest between X and Y occur only over time but we additionally include unanalysed (or nuisance) covariances between them to account for contemporaneous causal effects of ambiguous direction. In contrast, panel (b) in Figure 2, which corresponds to the theoretical model in panel (b) of Figure 1, features contemporaneous structural paths $X_t \to Y_t$. This means there are no longer any unanalysed covariances, so the statistical model is nearly identical to the DAG.[2] Note here Y_1 is now endogenous (and has an error term) while it was exogenous (no error term) in panel (a). The models in both panels assume no serial correlation as indicated by the absence of paths between the error terms in different time periods. While, unlike a DAG, an SEM permits contemporaneous reciprocal causal effects, we suggest that the necessary non-recursive simultaneous equations be used only as a last resort, given their often tenuous theoretical grounding and in particular strong assumptions required for identification (Wooldridge 2010).[3]

[1]SEMs include a measurement component, which specifies measurement models linking manifest indicators and latent variables, and a structural component, which specifies the relationships between latent variables. For now, we are exclusively referring to the structural component of an SEM. Those include bi-directional arrows, which depict the estimation of a covariance between two variables (or their error terms if they are endogenous), and one-directional arrows, which indicate the estimation of directional association based on regression modeling.

[2]Provided an instrument for X_t, one could add a covariance between ζ_t and ϵ_t to adjust for contemporaneous reverse causality.

[3]Plausible instrumental variables are notoriously difficult to find and non-recursive simultaneous equations require instruments for *each* endogenous variable of interest.

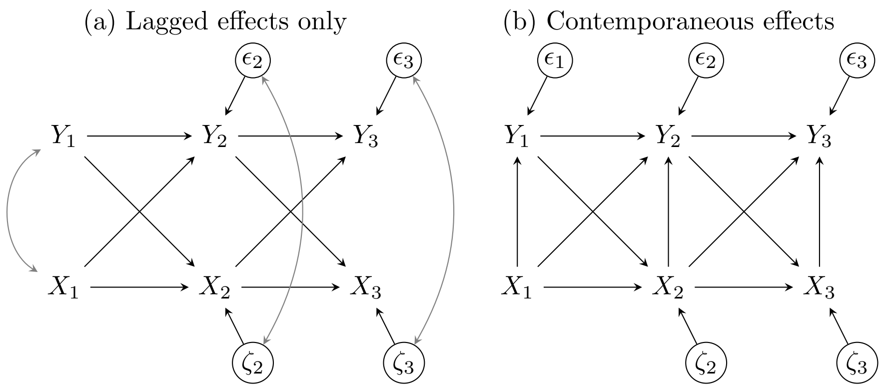

Figure 2: SEM path diagram representations of three-wave CLPM. Panel (a) features only lagged effects but permits ambiguous contemporaneous covariances between X and Y. Panel (b) features contemporaneous effects of X on Y.

When is a CLPM needed?

A cross-lagged panel model is only needed when reciprocal effects over time are presumed in the theoretical model. When the researcher can theoretically assume no effects of the dependent variable on the independent variable over time, other statistical models will be more appropriate than a cross-lagged panel model (see Imai and Kim 2019). For example, research focused on the effects of a single-event exposure—a common topic in life-course criminology and the role of turning points, such as the long-term effects of school drop-out or marriage on offending (Sampson and Laub 2016; Nguyen and Loughran 2018)—would not benefit from cross-lagged panel models.

Instead, reciprocal effects over time are most often a threat when the exposure is a continuous process that happens repeatedly. There are two situations in which reciprocal effects could emerge from theoretical models, thus highlighting the need of cross-lagged panel models. We name them *reverse causality* and *theoretical reciprocality.*

Reverse causality refers to situations in which researchers are primarily interested in assessing the effects of an exposure on an outcome but are concerned about the possibility that the outcome also influences future exposures. From our empirical example, assessing the effects of employment on crime involvement is of particular theoretical and policy relevance (Sampson and Laub 1993; Crutchfield 2014); in order to assess that relationship, however, one needs to take into account that offending behavior may also have an effect on future employment, even though examining prospects of job loss following a criminal arrest or behavior is not of substantive interest. This is a classic endogeneity problem, in which not accounting for reverse causality introduces bias. This has been a long concern of econometricians—e.g., the Arellano-Bond estimator, which is essentially a cross-lagged panel model, was developed specifically to address endogeneity in dynamic economic relationships (Arellano and Bond 1991; Moral-Benito, Allison, and Williams 2019). When dealing with reverse causality, researchers have a predefined exposure and an outcome of interest and simply aiming to control away the effects of the outcome on the exposure over time.

What we call theoretical reciprocality, on the other hand, refers to situations in which the directionality of the effects is ambiguous. This could due to, for instance, competing theories suggesting opposing pathways between two variables (Sturgis et al. 2004). From our empirical example, some researchers argue that perceived disorder leads to increases in fear of crime (Kelling, Wilson, et al. 1982) whereas others approach this relationship from the opposite direction, arguing that fear of crime leads local residents to view disorderly signs in the neighborhood as problematic (Sampson and Raudenbush 2004; Jackson 2004). This theoretical ambiguity implies that reciprocality is an empirical problem—a question that can be adjudicated by theory testers (Brunton-Smith 2011). When dealing with theoretical reciprocality, researchers often do not have a predefined exposure and an outcome of interest and instead aim to assess both directions.

In both cases, the theoretical model will highlight the threat of reciprocal relationships. A DAG will not distinguish between reciprocal effects due to theoretical reciprocality or due to reverse causality; this distinction is purely conceptual and should be dictated by the research question. Yet, we sustain that this is an important distinction. While in some cases the empirical solution will be similar, in other situations dealing with theoretical reciprocality or reverse causality may lead to different statistical estimators—we expand on this point below.

Panel Data

For the discussion that follows, let us assume that in both example cases we have three waves of panel data with yearly waves of observation and continuous measures of our concepts. In our first example—the effect of employment on offending and attendant reverse causality—employment is measured as the proportion of weeks spent employed in the last year and offending is a self-reported count of offenses committed in the past year. Both measures are taken at survey time but describe conditions or events occurring some time over the span of the year prior to each survey wave. In our second example—the theoretical reciprocal effects of perceived disorder and fear of crime—both perceived disorder and fear of crime are measured at survey time. That is, we observe respondents' "current" perceptions of disorder in their neighborhood and their "current" fear of crime in that neighborhood. To simplify examples, we will assume this is measured using a validated battery of indicators that produce a proportional score from 0 (no perceived disorder or fear) to 1 (maximum possibly disorder and fear).

Theory is recursive and operates at some defined pace—often discrete but sometimes continuous—but practical limitations of measurement commonly result in data collected in waves. The widths of time intervals between waves are usually determined by pragmatic considerations rather than theory, and thus the timing of observations rarely corresponds to the timing we expect in our theories. Often timing is inconsistent across variables as well, e.g., measures describing activities in the past year sit alongside measures of present perceptions. More generally, we can imagine the discordance between our theoretical model and our empirical data collection as a missing data or latent variable problem as illustrated in Figure 3. As noted, in the real world, time is continuous, so one might imagine there exists an infinite number of these “missing waves” between observations. Whether explicit or not, researchers always make assumption about the way variables and relationships operate within these unobserved intervals. Theory should and sometimes even does provide a guide to what assumptions are reasonable to make.

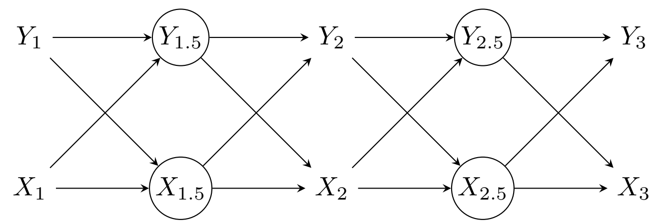

Figure 3: A theoretical model of cross-lagged effects with incompletely observed periods. Circles variables are latent.

When observation periods correspond in some way to the speed at which causal processes are believed to operate, theory may imply that you can rule out reciprocal effects or reverse causality. For example, if employment were measured as “Are you presently employed?” we could assume employment can be affected by past-year offending but employment cannot affect past-year offending. In neither of our present examples, however, do our empirical observation periods align with the theoretical causal process. Here we focus on cases where the processes we are interested occur more quickly than data are recorded[4]—that is, we must model a micro-level process using a macro-level aggregation. The reverse case—when recording occurs more quickly than the causal process—is easily addressed by aggregation or temporal lags.[5] While we restrict ourselves here to the treating time as discrete in both theory and analyses as this is the dominant approach in this literature, it is also possible to model cross-lagged relationships using continuous time dynamic models (Voelkle et al. 2019).

Temporality and composite variables

Before proceeding to the statistical model, it is necessary to discuss problems that can result from measurement strategies common in panel studies. For this purpose, we will distinguish between two types of measure, which we will call *instantaneous measures* and *period measures*. An instantaneous measure is one where the measured values correspond to some single instant

[4]Mulder et al. (2025) also note the possibility of “low process coverage” that occurs when observation intervals that are too long combine with time aggregated variables to make it so an insufficient amount of the underlying continuous process is captured by the data.

[5]This is similar to using using lags of variables from neighbors in spatial models to address the possibility that processes do not perfectly on to the chosen geographic units.

in time (or at least an exceedingly short period). Our perceived disorder and fear of crime variables are examples of instantaneous measures, as they would typically be measured using respondent perceptions at survey time. This situation corresponds roughly to that depicted in Figure 3 and be addressed with conventional approaches to causal inference with DAGs.[6]

In contrast, a period measure is one where the measured value corresponds to some deterministic sum (or other aggregate) calculated over a specified period of time. A count or variety score of offenses committed over the past year and proportion of time spent employed in the past year are both examples of period measures. When our observation periods do not correspond to the timing of our underlying theoretical causal model, these period measures become what we will call *composite period variables*. They are a composite because the measured value is a deterministic sum of values that would be obtained by dividing the observation period up into smaller intervals and aggregating across them.[7] This situation, appearing in Figure 4, produces challenges that are not well recognized in the literature and sometimes difficult to address (see Berrie et al. 2025).

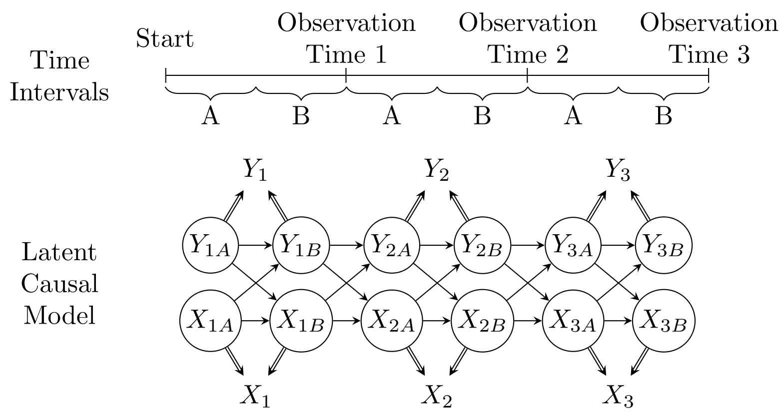

Figure 4: Reciprocality due to composite period variable. Circled variables are latent. Double-arrows indicate observed variables are composite variables determined by their parents

Figure 4 illustrates a case where we are interested in the relationship between two period variables A and B. The recall period for data collection encompasses 1 unit of time (e.g., an entire year) but the causal process relating the two period variables operates over smaller intervals (e.g., six months) denoted A and B. This is represented by cross-lagged effects between latent X and Y variables occurring across each smaller interval. Because data were only collected at each observation time, the measured values represent a composite period variable combining the values from periods A and B. For example, if Y_1 is the number of

[6]Note that the theoretical causal variable of interest may instead be this variable's value in some prior period; the available instantaneous measure thus represents a proxy for the past variable. The measure may be a poor proxy if the underlying variable varies significantly over time.

[7]See also Mulder et al. for a discussion of "time aggregation" where the measurement represents some average over the time period *as recalled by respondents*.

self-reported criminal offenses in a year, it would be the simple sum of the offenses in each six month period $Y_{1A} + Y_{1B}$. While we may know this sum, without additional information, we are uncertain of the value of either Y_{1A} or Y_{1B} (unless $Y_1 = 0$). Borrowing the notation of Berrie et al. (2025), the double-arrows indicate that Y_t and X_t are *deterministic* variables, i.e., they are fully determined by *some* combination of the underlying latent parent variables, even if we do not know their relative contribution[8]. Note that despite $X_t = X_{ta} + X_{tb}$, X_t is not a compositional variable as defined by Berrie et al. (2025), because the parents X_{tA} and X_{tB} are not simultaneously determined. Rather, X_t is a composite derived variable where the parents are unobservable or latent.[9] Composite measures present unique challenges for causal inference (see Berrie et al. 2025), of which we will highligh two: time-varying confounding and consistency violations. Time-varying confounding can be a serious problem as Berrie et al. (2025, 475) note:

> …if the parent variables occur at different moments in time, it is possible they will have different relationships with the supposed confounders. Indeed, it is possible that a confounder for one parent variable may be a mediator for another, leaving no means to appropriately adjust for confounding without also blocking part of the true causal effect. Identifying the causal effect of a composite derived variable, as with any analysis of multiple exposures, therefore requires no time-varying confounding.

Consistency refers to the assumption required for making causal inferences that treatments are well-defined, in particular that only one unique form of treatment exists at any given measured level of the exposure variable. This may be violated by composite variables because one can arrive at the same value of the observed child treatment variable through multiple combinations of the parent variables; i.e., someone who worked half of the past year could have worked only first half, only second half, or some mix of those extremes. This can produce composite variable bias, because the estimated average causal effect of the composite variable will be a mean of the average causal effects of the parent variables that is proportionally weighted toward the parent(s) with larger variance(s).

Estimands

For our first example—the effect of employment on offending which is threatened by reverse causality—we expect that there is a delay between when someone is caught offending and when they could be fired from their job but either employment or offending could have occurred anywhere throughout the year covered by the measurement. If we decreased gaps between waves—perhaps to daily or weekly observations—or could obtain a plausible instrumental

[8] If Y_{tA} and/or Y_{tB} contain measurement error, this full measurement error is propagated into the composite variable as well.

[9] If you have a strong theoretical reason to assume a variable affects one parent but is exogenous to the other, you could use it as an instrument to isolate the parent you are interested in, e.g., a randomized employment program affecting only the second half of the year.

variable we could convincingly rule out reverse causality, at least for the contemporaneous effect of employment on offending, e.g., the "involvement" effect in Hirschi's control theory (Hirschi 1969). With the available data, however, we can only consider employment effects occurring over a considerable amount of time, such as the effect of "commitment" to conventional lines of action (Hirschi 1969). Accordingly, we specify our estimand as the effect of proportion of time employed in the prior year (x_{t-1}) on offending in the following year (y_t). For our second example—the theoretically reciprocal effects of fear of crime and disorder perception—we expect that both processes of updating could occur quite rapidly—possibly more quickly, even, than could be captured with *daily* data. Convincingly determining causal ordering would require biophysical measurements in a laboratory, an approach which would come at a cost of construct and external validity. But perhaps we are interested in how this updating process produces lasting changes in how the environment is perceived and if one side of the process operates more strongly than the other. Based on this, we will specify two estimands of interest: the effect of disorder perceptions (fear perceptions) measured in the prior year (x_{t-1}) on fear perceptions (disorder perceptions) measured in the present year (y_t). Thus, in both cases, we are confident about the the direction of the lagged and cross-lagged relationships we expect to observe in our available data but uncertain about the direction (or even existence) of contemporaneous relationships.

Statistical Model

Once the theoretical model and estimands have been established, one can proceed to developing a statistical model that accounts for the constraints from the research design—including ambiguity about causal directions. Here, path diagrams from the SEM tradition become useful. SEM diagrams encode both causal relations between variables—indicated by single-headed arrows pointing from cause to effect—and unanalyzed, unexplained, or "nuisance" associations—indicated by double-headed arrows pointing between two variables. That is, one explicitly distinguishes between causal relationships and ambiguous associations. Variables with any causal path leading into them are termed "endogenous" variables and those with no causal path leading into them are termed "exogenous". SEM diagrams depart from most DAGs by depicting the residuals or unexplained variation of endogenous variables as latent variables causally affecting the observed variable.

Our recommendation is that the causal part of the SEM diagram should be recursive, corresponding as closely as possible to the theoretical model. Reciprocal simultaneous effects should generally be avoided as they require strong identifying assumptions, e.g., valid and relevant instrumental variables (Green 2003; Wooldridge 2010). When the timing of observation results in ambiguity regarding the causal direction within any given time period—such as when periods $t = 1.5$ and $t = 2.5$ were not observed in Fig XXX—we encode this ambiguity as an unanalyzed covariance between the variables in question. When the timing of observations closely corresponds to theory, these covariances are unnecessary and the resulting statistical model similarly corresponds directly to the theoretical model.

For both examples, following these guidelines leads us to the "classic" three-wave CLPM depicted in FIGXX. The causal relations consist of two sets of relationships: a) lagged (autoregressive) effects of variables in one period on their value in the next period, e.g., $x_t \rightarrow x_{t+1}$, and b) cross-lagged effects of variables in one period on the other variable in the next period, e.g., $x_t \rightarrow y_{t+1}$. The non-causal relations consist of two sets of covariances: (a) between the exogenous x_1 and y_1, which merely permit the predictors to be correlated as in any typical regression model, and (b) between the residuals of the endogenous variables (ζ_t and ϵ_t)[10]. The residual covariances account for the ambiguous contemporaneous association between x_t and y_t but cannot tell us the direction of any causal effect between the two variables.

In principle, the classic CLPM permits identification of the cross-lagged effects under conventional assumptions common to most causal inference approaches (see Morgan & Winship 2014; Wooldridge 2010) and in paticular mediation analysis, e.g. sequential ignorability . In application, researchers using the classic CLPM may easily fall prey to a number of common issues that, if unaddressed, result in misleading inferences. Rather than providing an exhaustive list of problems that can arise with CLPMs, we focus on the three key issues we believe are the most important for researchers in developmental and life course criminology to be aware of due to the frequency with which they will be encountered and the severity of their effect on inferences: (a) unobserved heterogeneity, (b) improperly specified temporal order, (c) insufficient inter-temporal variation.

Issue 2: Unobserved heterogeneity

One of the biggest advantages of drawing on longitudinal data is the ability to focus exclusively on within-unit variation (i.e., change over time) and control away any stable, trait-like differences between units (Imai and Kim 2019; Allison, Williams, and Moral-Benito 2017). For example, by only drawing on within-unit variation, the family of regression models known as fixed effects models ensure that any confounding bias emerging from time-constant confounders is immediately accounted for (although see Imai and Kim 2019). In the traditional cross-lagged panel model, autoregressive paths are supposed to serve as a proxy for fixed effects, capturing all stable heterogeneity, thus accounting for time-invariant traits and ensuring that cross-lagged paths reflect effects on change over time.

However, autoregressive parameters alone are not sufficient to remove all unobserved stable heterogeneity (Morgan and Winship 2014). Any stable between-person unit—that therefore influences both an outcome Y_t and a lagged outcome Y_{t-1}—creates a spurious association between the two variables that the model incorrectly attributes to within-unit dynamics. This inflated association then contaminates the the cross-lagged coefficients (Hamaker, Kuiper, and Grasman 2015), which no longer reflects *only* effects on change over time. This implies that the standard CLPM conflates within-unit change with between-unit differences, and cross-lagged estimates can reflect both sources of variation.

[10]Omitting the residual covariances yields the oft-maligned cross-lagged correlations model.

If autoregressive parameters alone do little to adjust for all unobserved time-stable confounders, one might contemplate including some fixed-effects estimator [e.g., within-unit demeaning; see Allison (2009)]. However, this strategy introduces other problems. Imai and Kim (2019) showed that unit fixed effects only consist of a non-parametric solution to adjusting for all time-stable heterogeneity under the assumption of no reverse causality or causal dynamics—that is, the assumption that past outcome does not affect current treatment, and that past treatment does not affect current outcome. Such dynamic relationships are precisely the issue that CLPMs attempt to address, so they cannot be assumed away. In addition, including fixed effects alongside autoregressive parameters in a dynamic panel model can lead to the well-known *Nickell bias*: estimates are downward biased with few periods (i.e., small T) because of the connection between the lagged outcome and the fixed effects (Wooldridge 2010). In the econometrics literature, this is usually solved with the Arellano-Bond estimator (Arellano and Bond 1991), which removes Nickel bias and adjusts for unobserved stable heterogeneity—but it yields biased estimates with small samples and when the autoregressive parameter is near 1 (i.e., highly stable over time).

Stable unobserved heterogeneity is therefore an important issue. The standard CLPM relies on autoregressive parameters to capture temporal stability, but this does not properly ensures that only within-unit variation is taken into account. Any unmodeled time-stable confounder can bias the cross-lagged parameters, which end up conflating within-unit change with between-unit differences. The Arellano-Bond estimator of the dynamic panel model can solve this issue by including fixed effects, but it has other limitations.

Solutions

While the traditional CLPM cannot properly account for unobserved stable heterogeneity, some relatively recent extensions of the CLPM yield robust estimators that draw only on within-unit variation and effectively control for all unobserved time-stable traits (Leszczensky and Wolbring 2019). Specifically, Allison et al.'s (2017) maximum likelihood structural equation fixed-effects dynamic panel method (ML-SEM) and Hamaker et al.'s (2015) random intercept cross-lagged panel model (RI-CLPM) model reciprocality and account for unobserved, time-stable features that impact both Y_t and X_t.

Fixed-effects dynamic panel model with maximum likelihood

In ML-SEM (Allison, Williams, and Moral-Benito 2017), a standard dynamic panel model (with a lagged dependent variable) is recast as a structural equation model that explicitly partials out all unobserved stable heterogeneity. In practice, latent random intercept factor (often denoted α) is introduced with a factor loading constrained to 1 on every observed Y_t. This latent random intercept factor is the same as the one often included in latent growth curve models (e.g., a random intercept reflecting the starting point of latent trajectories) and represents all between-unit differences. By fixing the factor loadings of the random intercept to

1, the model effectively "demeans" each unit's scores and thus conditions out all time-constant heterogeneity.

Crucially, this latent random intercept factor is allowed to freely covary with all the time-varying exogenous predictors (including any predetermined regressors and the initial outcome Y_1). This model is exactly a correlated-random-effects model (Mundlak 1978): by letting α correlate without restriction with all X's, the SEM mimics a fixed-effects specification (the contrast being that a classical random-effects model would force α to be independent of X). As Allison, Williams, and Moral-Benito (2017) note, allowing α to covary with X's enforces a fixed-effects model that accounts for all unobserved, stable confounders. This means that the ML-SEM assumes a key outcome (e.g., Y) and a key predictor (e.g., X) of interest, and the whole model is set in such a way so as to find effects of the predictor on the outcome (e.g., cross-lagged and/or contemporaneous effects of X on Y) while accounting for a potential covariance between past outcome (Y_{t-1}) and current predictor (X_t). If cross-lagged or contemporaneous effects of the outcome Y (e.g., Y_{t-1} or Y_t) on the predictor X_t are also of substantive interest, one needs to fit a separate ML-SEM with that specification.

In implementation, the entire system of autoregressive and cross-lagged equations is cast as a single SEM. For example, each outcome (Y_t) is regressed on its lagged value (Y_{t-1}), on a contemporaneous (X_t) and/or a lagged value of the predictor (X_{t-1}), and on a latent random intercept factor α with a constrained loading—here specified as a fixed effect. Figure 5 illustrates the ML-SEM with four waves.[11] α represents the latent random intercept factor as is reflected by all observations Y_t (except for the initial observation $Y_{t=1}$) with factor loadings constrained to 1, and all X_t are allowed to correlate with α (and with each other and with $Y_{t=1}$). In this setup, the first observation $Y_{t=1}$ is treated like any exogenous covariate. Importantly, observed time-varying regressors (X_t) are specified as predetermined—that is, they are allowed to correlate with past residuals (i.e., with $\epsilon_{y,t-1}$), reflecting the idea that predictors may be influenced by prior outcomes. The model is equivalent to the Arellano-Bond method estimated with the generalized method of moments (Arellano and Bond 1991), but it requires no stationarity assumption on initial conditions (given that initial observations $Y_{t=1}$ and $X_{t=1}$ are simply treated as additional exogenous predictors). The ML-SEM also expands upon the Arellano-Bond method as it can handle missing data with full information maximum likelihood, it can easily extend to multiple indicators, and it can incorporate time-constant covariates and different timing effects. The ML-SEM model is a standard SEM with autoregressive and cross-lagged parameter using only within-unit variation and a latent random intercept α factor soaking up all between-unit variation.

Random intercept cross-lagged panel model

The other alternative to the traditional cross-lagged panel model that actually controls for unobserved stable heterogeneity is Hamaker et al.'s (2015) random intercept cross-lagged panel

[11]The ML-SEM requires at least four waves of data to be implemented.

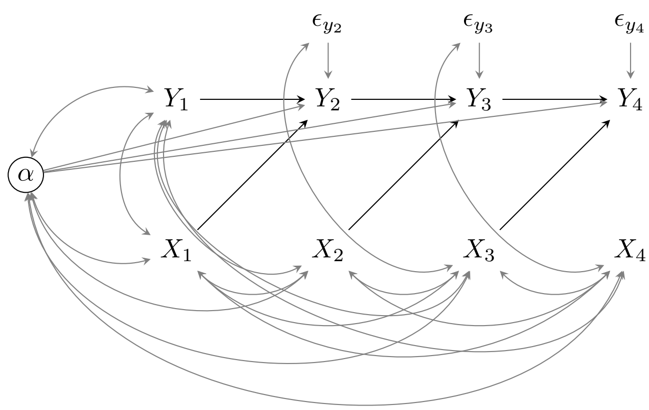

Figure 5: Maximum likelihood structural equation fixed-effects dynamic panel model (Allison et al., 2017)

model (RI-CLPM). To ensure that the model only takes into account change over time, the RI-CLPM explicitly models within-unit and between-unit variances as separate attributes. Similar to Allison et al.'s (2017) ML-SEM, the first step involves estimating a latent variable that captures all stable variation between units—again drawing on the latent growth curve model specification, this latent random intercept factor is defined by all observations Y_t over time with factor loadings contrained to 1, essentially creating a random intercept for Y that varies between units.

While the ML-SEM involves allowing the latent random intercept factor to covary with pretedermined regressors—which implies treating it as a fixed effect that captures all unobserved stable heterogeneity—the RI-CLPM instead repeats the process and creates another latent random intercept factor defined by all observations X_t over time, thus creating another latent random intercept for X that varies between units. Treating such unit-specific random intercepts as random variables (as opposed to fixed parameters) usually implies that they do not capture all unobserved stable heterogeneity. However, in the RI-CLPM, the two latent random intercept factors for Y_t and X_t (represented by α_Y and α_X in Figure 6) are allowed to covary, which ensures that they capture all unobserved stable heterogeneity across both variables. Figure 6 depicts the the RI-CLPM.

To further ensure that only within-unit variation is taken into account, the RI-CLPM relies on a set of other latent variables. For each variable Y_t and X_t, a new latent variable is estimated with a factor loading constrained to 1. These latent variables are represented in the diagram in Figure 6 as W_{Y_t} and W_{X_t}. This implies that every single observed variable Y_t and X_t is defined by two latent variables: one capturing all differences between units (i.e., α_Y and α_X), and one capturing everything else, i.e., within-unit differences (i.e., W_{Y_t} and W_{X_t}). Autoregressive and cross-lagged parameters are then included linking W_{Y_t} and W_{X_t}—

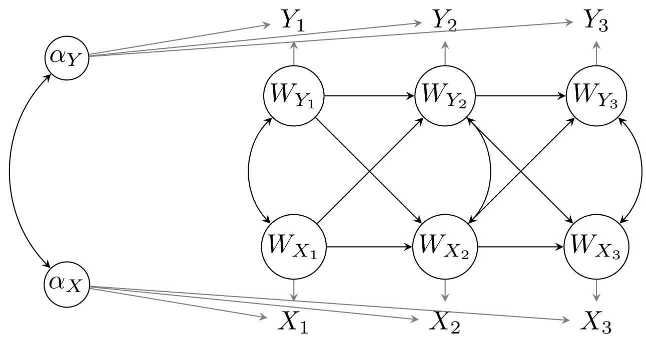

Figure 6: Random Intercept Cross-Lagged Panel Model (Hamaker et al., 2015)

resembling the traditional cross-lagged panel model, but exclusively focused on within-unit variation over time. By allowing both latent random intercept factors to covary, the RI-CLPM ensures that all unobserved stable heterogeneity is accounted for; and by estimating autoregressive and cross-lagged effects on latent variables capturing only within-unit variation, the model ensures that cross-lagged estimates reflect change over time. The RI-CLPM can easily be extended to other common features in SEMs, such as multiple indicators, covariate adjustment, and multiple-group analysis (Mulder and Hamaker 2021).

Issue 1: Temporal Order

Temporal order presents two key issues for studying reciprocal effects. The first is that researchers may fail to consider time in their theoretical model to begin with. This may result in choosing an estimator before specifying—or perhaps without even knowing—their estimand. If yearly panel data are available, they simply estimate the effect of some past-year measure on a present year measure. This may also result in researchers assuming that if they fail to detect an effect in one lag structure it implies effects do not exist at other temporal lags. Researchers may instead assume there exists one "true causal effect" between two variables which is ambiguous in timing and set out to use a model to determine at what timing this effect exists. This raises the question of whether different causal timings could possibly correspond to comparable or even plausible processes of interest, and secondarily whether any model is likely to be able to identify multiple causal timings simultaneously.

Improper specifications of temporal order may have serious consequences for estimation. Simulations specifying a causal effect that exists at only one particular time lag reveal that incorrect specifications of temporal lags in statistical models with unobserved effects (i.e., fixed effects) can produce estimates with severe bias including flipped signs. As Vaisey & Miles (2017) document, if Equation 1 is the true reduced form model of interest (x has only a contemporaneous effect) and you instead specify Equation 2 in your statistical model (x with only a lagged effect), then $E(\beta^*) = -0.5\beta$, i.e., “a positive contemporaneous effect of x gets transformed into an artifactually negative lagged effect” Allison (2022).

$$y_{ti} = \beta x_t + \alpha_i + e_{it} \tag{1}$$
$$y_{ti} = \beta^* x_{t-1} + \alpha_i + e_{it} \tag{2}$$

This is possibly the least appreciated and most serious common issue facing CLPMs as it may produce misleading results that are interpreted as evidence not just against the theoretical proposition being tested by even of its opposite; to use one of our examples, this could result in research finding employment increases future offending even when it has only a prevantative contemporaneous effect.[12] Research that finds unexpectedly negative estimates from CLPMs when theory suggests positive effects may be the result of this bias.

Solutions

As we have stated earlier, there is no “one true causal effect” but rather a number of causal effects corresponding to different estimands researchers choose. In this regard, what is important is specifying the correct lag structure *for the chosen estimand*. A null effect for a particular time lag may be compatible with a non-null effect at a different time period; a study that finds no effect of an additional pint of beer on decision-making with a lag of one day provides little information on the effect at one hour. It is unlikely a researcher has no theoretical expectations about what causal timings are of interest. If this is the case, however, the researcher is not prepared to engage in a causal analysis. Determining which lags are of theoretical interest is a question for literature reviews and exploratory analyses.

The prior equations specify different estimands of interest, though both make the strong assumption there is no reverse causality between x_t and y_t. If the estimand is the contemporaneous effect of x_t on y_t and there is concern about reverse causality, some auxiliary information is necessary such as an instrumental variable. Without a theoretical justification otherwise, it is reasonable to default to including a cross-lagged effect as well ($x_{t-1} \rightarrow y_t$). If also accounting for unobserved heterogeneity, this necessitates use of the models above which address Nickell bias occurring in dynamic panel models with unobserved effects. If the estimand of interest is the lagged effect of x_{t-1} on y_t however, then reverse causality could be addressed in a cross-lagged model with a covariance between each x_t and y_t to address this nuisance reciprocality that may bias the estimate of interest Allison (2022). This is a more robust specification if the parameter of interest is βx_{t-1} above, but it cannot estimate βx_t or adjudicate between different the directions of contemporaneous effects.

[12] Note that this is not the result of signs differing between estimands but rather a biased estimate resulting from model misspecification in equation 2.

Issue 3: Inter-temporal variation

Many of the focal causes and effects in developmental and life course criminology are relatively stable over time in some sense, such as psychological constructs (e.g., self-control) and life-course statuses (e.g., marriage). By definition, in cross-lagged panel models the outcomes serve as predictors of later outcomes—both as lags (e.g., $Y_{t-1} \rightarrow Y_t$) and cross-lags (e.g., $Y_{t-1} \rightarrow X_t$). If an outcome is relatively stable over time—that is, strongly dependent on its past values—there may be little remaining to be explained by the predictors of interest after accounting for its prior value. Put another way, as the correlation between past and present values increases ($\rho(Y_t, Y_{t-1}) \rightarrow 1$) the remaining variance that can be explained by other covariates shrinks ($Var(Y_t|Y_{t-1}) \rightarrow 0$).[13] This is an imperfect multicollinearity problem: estimates may remain unbiased and consistent but imprecise and even unstable or empirically underidentified when the issue is extreme. This is exacerbated in estimators with unobserved effects as relatively stable outcomes may be largely explained by the time-stable fixed effects or correlated random intercepts.[14] These issues are common in developmental studies of individuals individuals but also commonly found in other criminological literatures using panel data (see Lanfear et al. 2020). Researchers may also be interested in how relatively stable factors influence faster moving processes, i.e., moderation. Low inter-temporal variation is not inherently a problem for these research questions as interaction terms between time-stable and time-variant factors will be time-variant, though interaction terms still make strong demands of the data (e.g., Gelman 2023).

Solutions

Ideally, researchers should account for the possibility of low inter-temporal variation during design. Researchers interested in the effects of changes in relatively stable outcomes can obviously opt to collect data over long time periods. If doing so is cost prohibitive or could prevent answering a timely research question, there are at least two other options. First, if the normally time-stable factor is affected by external factors—including those that may be manipulated by researchers—then researchers can collect data on individuals exposed to shocks in these factors—or embed an intervention in their study to manipulate the external factor. This then opens the possibility of using a instrumental variables estimator to estimate a local average treatment effect. Second, even if the time stable factor is not responsive to exogenous shocks, the factor may be less stable among some subpopulations. In this case,

[13] Note that if the variables are measured with error, what is relevant is the underlying correlation between the error-free latent outcomes; when measurement error is present, it becomes an increasingly large proportion of the conditional variance as the true correlation increases. This may result in even more imprecise estimates while also attenuating estimates as the outcomes also serve as regressors. This may be addressed using measurement models or errors-in-variables estimators.

[14] One may also include time-invariant covariates in unobserved effects models using a hybrid random approach, however the estimates will correspond to between-individual effects which may differ from effects of within-individual changes and must be assumed to be uncorrelated with unobserved effects.

researchers might oversample the more rapidly changing subpopulation to maximize variation, while remaining aware that the effect identified in that group may differ from that in the larger stable population.

Researchers using already-collected data may consider a number of approaches. First, they may consider estimands corresponding to shorter time aggregations or longer time intervals. Disaggregation may help as low variation at a larger temporal unit (e.g., a year) may mask higher variation in smaller units (e.g., a month). However, unless such an estimator is intractable with many time periods, researchers should default to disaggregated analysis as longer-term causal effects can always be captured using models with multiple lags of shorter periods. In extreme situations, researchers may consider skipping waves if it makes sense theoretically (e.g., skipping a mid-year wave when considering an annual process) or they are facing serious empirical underidentification problems related to a particular wave (e.g., a very short observation interval resulted in no change between waves). In any case, researchers are again advised not to substitute a good question that is difficult to answer (i.e., a theoretically-motivated estimand) for a bad question that is easy to answer (i.e., anything else).

General Advice

Our key recommendations for researchers studying potentially reciprocal relationships with repeated observations may be summarized in three points.

First, it is important to separate theory problems from estimation problems. A clear theoretical model should be constructed first based on deep knowledge of the substantive research area and the data generation process. From this, researchers should then determine what, if any, theoretically-relevant estimand can plausibly be estimated using the available data. With a theory and estimand in hand, researchers should then choose an appropriate estimator. Readers may also apply this logic when reading research, asking if the stated or implied estimand one of substantive interest or one that corresponds to a plausible mechanism. If not, a strong estimation strategy cannot make up for a theoretical deficit.

Second, researchers should think carefully whether the data they have are suitable for answering their questions to begin with—that is, they should be prepared to stop after the second step above. Conventional longitudinal panel survey data are better equipped to test slow and lasting effects proposed by developmental and life course theories (e.g., trajectories and turning points) than rapid or transient processes from cognitive and interactional theories (e.g., decision-making and group processes). For testing theories that connect long-term and short-term processes, researchers should consider incorporating other forms of data collection into panel studies—e.g., experiments (Farrington, Loeber, & Welsh 2009) or momentary assessments (Browning et al. 2024). Panel data with narrow observation intervals are sometimes also poorly suited for testing slow processes; high intertemporal correlations and proportions of variance explained may be signals there is insufficient change over time to produce precise estimates. Researchers and readers should also consider if they are using composite period variables and how this may impact the validity of their designs and estimates; these variables may correspond to unusual estimands, violate consistency assumptions for making causal inferences, and make it difficult to plausibly address reverse causality.

Third, researchers should default to robust estimators. If the estimand is a lagged effect, include a contemporaneous covariance to adjust for potential reverse causality. If the treatment and outcome of interest vary (sufficiently) over time, adjust for unobserved heterogeneity using ML-SEM or RI-CLPM. If the estimand is a contemporaneous effect (or the effect of a stable trait) or there is insufficient variation to use these robust estimators, researchers should state outright the strong assumptions they are making—which, importantly, may still be the best that can be done to answer an important question until better data and designs are available. When evaluating research, readers should ask whether the chosen estimator is consistent with the theoretical model and robust to common issues we describe. In particular, estimates opposite in sign of theoretical expectations should invite skepticism and scrutiny.

Finally, while this work focuses on issues specific to studies positing reciprocal effects, researchers and readers should also be attentive to key issues that apply to all quantitative studies testing theories such as plausibly establishing (conditional) ignorability, consistency, and absence of interference (Morgan & Winship 2014; Rubin 1986). In longitudinal contexts, this includes considering dynamic effects (i.e., lagged outcomes) and time-varying confounding. Where these are difficult to address, exogenous shocks, field interventions, and embedded lab experiments may provide plausible identification strategies. General training in principles for causal inference thus serve as a foundation for strong longitudinal research designs. Fortunately, barriers to entry for developing these skills have rapidly eroded with the proliferation of accessible introductory materials (e.g., Morgan & Winship 2014; McElreath 2020; Huntington-Klein 2025; Hernán & Robins 2020; Imai & Llaudet 2022).

Conclusion

- What to do for a strong conclusion? Revisit each key idea?
  - Know yout theory and use it to specify an estimand
  - Thinking carefully about temporal order is important for both designing and interpreting research.
    - For interpretation, readers should longitudinal studies with unexpectedly negative signs deserve additional scrutiny.
  - Unobserved heterogeneity
  - Intertemporal variation
- Or do something else? Pontificate in some way? Kind of feel prior section is the pontification...

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